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Microelectronics package comprising a package-on-package (PoP) architecture with inkjet barrier material for controlling bondline thickness and PoP adhesive keep out zone

Abstract

Embodiments disclosed herein include electronic packages. In an embodiment, the electronic package comprises a first package, wherein the first package comprises, a first package substrate, a first die over the first package substrate, a first mold layer over the first package substrate and around the first die, and a plurality of through mold interconnects (TMIs) through the first mold layer. The electronic package may further comprise a second package electrically coupled the first package by the TMIs, wherein the second package comprises a second package substrate, a second die over the second package substrate, and a solder resist over a surface of the second package substrate opposite from the second die. In an embodiment, the electronic package may also comprise a barrier between the first package and the second package.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/374,587, filed Sep. 28, 2023, which is a continuation of pending U.S. patent application Ser. No. 18/372,542, filed Sep. 25, 2023, which is a continuation of pending U.S. patent application Ser. No. 16/557,784, filed Aug. 30, 2019, the entire contents of which are hereby incorporated by reference herein.

TECHNICAL FIELD

(1) Embodiments of the present disclosure relate to semiconductor devices, and more particularly to barriers disposed between a top package and a bottom package of a package-on-package (PoP) architecture.

BACKGROUND

(2) Thermal performance is a critical concern in many advanced packaging architectures. In particular, package-on-package (PoP) architectures require careful consideration of thermal performance since the underlying package (e.g., a system-on-chip (SoC) package) is entirely covered by the top package (e.g., a memory package). Accordingly, the path for thermal dissipation

from the SoC is up through the memory package. However, the thermal path has a high thermal resistance because there is typically an air gap between the top package and the bottom package. Currently, thermal interface materials are not included between the two packages because there is no way to control the flow of the material. That is, if a thermal interface material was disposed between the two packages, the thermal interface material would spread and bleed into the through mold interconnect (TMI) joints, which would lead to solder extrusion.

(3) Some solutions to the spreading of an interface material have been proposed. For example, solder resist dams may be used to confine a thermal interface material. However, the creation of the dam requires complex substrate processing, including lithography. This limits when the dam can be formed. That is, the dam must be formed during substrate manufacturing, and before any ball attach (BA) step, since the solder resist dam would impede BA paste printing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a cross-sectional illustration of an electronic package with a package-on-package (PoP) architecture that includes a barrier between a first package and a second package, in accordance with an embodiment.

(2) FIG. 1B is a cross-sectional illustration of an electronic package with a PoP architecture that includes a barrier that confines a thermal adhesive between a first package and a second package, in accordance with an embodiment.

(3) FIG. 1C is a cross-sectional illustration of an electronic package with a PoP architecture that includes a barrier between a first package and a second package, in accordance with an additional embodiment.

(4) FIG. 2A is a plan view illustration of a bottom surface of an electronic package that includes a barrier that is a ring, in accordance with an embodiment.

(5) FIG. 2B is a plan view illustration of a bottom surface of an electronic package that includes a first barrier ring and a second barrier ring, in accordance with an embodiment.

(6) FIG. 2C is a cross-sectional illustration of an electronic package that illustrates a profile of the barrier, in accordance with an embodiment.

(7) FIG. 3A is a cross-sectional illustration of a top package of a PoP architecture that includes a barrier, in accordance with an embodiment.

(8) FIG. 3B is a cross-sectional illustration of a top package of a PoP architecture that includes a first barrier and a second barrier, in accordance with an embodiment.

(9) FIG. 3C is a cross-sectional illustration of a bottom package of a PoP architecture that includes barrier, in accordance with an embodiment.

(10) FIG. 4A is a plan view illustration of an electronic package with a plurality of barriers, in accordance with an embodiment.

(11) FIG. 4B is a plan view illustration of an electronic package with a plurality of barriers, in accordance with an additional embodiment.

(12) FIG. 4C is a cross-sectional illustration of an electronic package with a plurality of barriers, in accordance with an additional embodiment.

(13) FIG. 5 is a cross-sectional illustration of an electronic system that includes a PoP architecture with a barrier between the top package and the bottom package, in accordance with an embodiment.

(14) FIG. 6 is a schematic of a computing device built in accordance with an embodiment.

EMBODIMENTS OF THE PRESENT DISCLOSURE

(15) Described herein are electronic packages with barriers disposed between a top package and a bottom package of a package-on-package (PoP) architecture, in accordance with various

embodiments. In the following description, various aspects of the illustrative implementations will be described using terms commonly employed by those skilled in the art to convey the substance of their work to others skilled in the art. However, it will be apparent to those skilled in the art that the present invention may be practiced with only some of the described aspects. For purposes of explanation, specific numbers, materials and configurations are set forth in order to provide a thorough understanding of the illustrative implementations. However, it will be apparent to one skilled in the art that the present invention may be practiced without the specific details. In other instances, well-known features are omitted or simplified in order not to obscure the illustrative implementations.

(16) Various operations will be described as multiple discrete operations, in turn, in a manner that is most helpful in understanding the present invention, however, the order of description should not be construed to imply that these operations are necessarily order dependent. In particular, these operations need not be performed in the order of presentation.

(17) As noted above, thermal performance of PoP systems is limited due in part to the air gap between the bottom package and the top package. Filling the air gap with a thermal interface material has not been widely adopted because the deposited thermal interface material will flow into the through mold interconnect (TMI) region. Solder resist dams have also not been a practical solution.

(18) Accordingly, embodiments disclosed herein include a barrier that is disposed between the first package and the second package. The barrier may be disposed with a printing process (e.g., ink jet printing). As such, the barrier may be disposed over the first package and/or the second package at any point in the assembly. This allows for standard packages to be obtained from the supplier and later modified to include the barrier immediately prior to PoP assembly.

(19) In an embodiment, the barrier provides a boundary that prevents a thermal adhesive from spreading uncontrollably. Therefore, the air gap between the first package and the second package may be filled by a thermal adhesive material without the concern of the material spreading to the TMI region. This reduces the thermal resistance along the path from the first package to the second package, and therefore, improves performance of the system. The barrier may also provide a boundary that keeps flux from the TMI region out of the die region.

(20) In an embodiment, the barrier may also be used as a bondline thickness (BLT) control mechanism. Without a BLT control mechanism, process variation may result in reductions in yield. For example, lack of BLT control may result in solder bump bridging in the TMI region. Accordingly, precisely controlling the BLT with a barrier improves yield.

(21) Referring now to FIG. 1A, a cross-sectional illustration of an electronic package **100** is shown, in accordance with an embodiment. In an embodiment, the electronic package **100** has a PoP architecture. That is, the electronic package **100** may comprise a first package **140** (also referred to as a bottom package **140**) and a second package **120** (also referred to as a top package **120**). In some embodiments, the first package **140** may be a system-on-a-chip (SoC) package, and the second package **120** may be a memory package.

(22) In an embodiment, the first package **140** comprises a package substrate **141**. The package substrate **141** may include a plurality of organic buildup layers with conductive routing (e.g., traces, vias, pads, etc.). A first die **144** may be coupled to the package substrate **141**. For example, first level interconnects (FLIs) **146** or any other suitable interconnect architecture may be used to couple the first die **144** to the package substrate **141**. In some embodiments, the first package **140** may further comprise a plurality of second dies **145**. The second dies **145** may be coupled to the first die **144** by FLIs **146** or other suitable interconnects. In an embodiment, the first die **144** may include transistors fabricated at a first process node and the second dies **145** may include transistors fabricated at a second process node that is more advanced than the first process node. In an embodiment, capillary underfill (CUF) **147** may be disposed around the FLIs **146** of the first die **144** and the second dies **145**.

(23) In an embodiment, the first package **140** may comprise a first mold layer **142**. The first mold layer **142** is over the package substrate **141** and surrounds the first die **144**. In an embodiment, a second mold layer **148** may embed the second dies **145**. Other embodiments may include embedding the first die **144** and the second dies **145** with the first mold layer **142**. In an embodiment, through mold interconnects (TMIs) **143** may pass through the first mold layer **142**. The TMIs **143** may be electrically coupled to the first die **144** and the second dies **145** by conductive paths (not shown) through the package substrate **141**.

(24) In an embodiment, the second package **120** may be electrically coupled to the first package **140** by the TMIs **143**. For example, the TMIs **143** may pass through a solder resist layer **122** of the second package **120** and land on conductive pads (not shown) of a package substrate **121**. In an embodiment, the second package **120** may include one or more dies **123**. For example a pair of dies **123** are shown in FIG. 1A, but it is to be appreciated that any number of dies **123** may be used in accordance with various embodiments. The dies **123** may be stacked over each other in some embodiments. A die attach film (DAF) **124** may couple dies **123** together. In an embodiment, the dies **123** may be offset from each other to allow for wire bonds **126** to electrically couple the dies **123** to the package substrate **121**. The wire bonds **126** may be electrically coupled to the TMIs **143** by conductive routing (not shown) through the package substrate **121**. In an embodiment, a mold layer **125** may be used to embed the dies **123**.

(25) In an embodiment, the first package **140** is spaced away from the second package **120** by a spacing **S**. The spacing **S** represents the BLT between the first package **140** and the second package **120**. In an embodiment, the spacing **S** may be approximately 10 μm or greater. In a particular embodiment, the spacing **S** may be approximately 25 μm or greater. Control of the spacing **S** may be provided by a barrier **131**. The barrier **131** may be an ink barrier that is printed onto either the first package **140** or the second package **120**. In an embodiment, the barrier **131** comprises a polymeric material. For example, the barrier **131** may comprise acrylic polymers. Additional embodiments may also comprise TiO_2 particles. The barrier **131** may be a UV curable material. Accordingly, the barrier **131** may be printed and subsequently cured to provide mechanical rigidity needed to control the BLT.

(26) In an embodiment, the barrier **131** defines an air gap **136** between the first package **140** and the second package **120**. In an embodiment, the air gap **136** is positioned over the first die **144** and the second dies **145**. In a particular embodiment, the barrier **131** may contact the first package **140** along the first mold layer **142** or the second mold layer **148**. The barrier **131** may contact the second package **120** along the solder resist **122**. For example, a first surface **135** of the barrier may be in direct contact with the solder resist **122**, and a second surface **134** may be in direct contact with the mold layer **142**. In an embodiment, the first surface **135** is substantially flat, and the second surface **134** is rounded or domed. This is indicative of the barrier **131** being printed on the second package **120**, and then brought into contact with the first package **140** during assembly. In an embodiment, the air gap **136** may be entirely surrounded by the barrier **131**. That is, the barrier **131** may be a ring. Such a configuration prevents materials, such as flux from the TMIs **143**, from migrating towards the die region.

(27) Referring now to FIG. 1B, a cross-sectional illustration of an electronic package **100** is shown, in accordance with an additional embodiment. In an embodiment, the electronic package **100** in FIG. 1B may be substantially similar to the electronic package **100** in FIG. 1A, with the exception that the air gap **136** is filled with a thermal interface material. For example, a thermal adhesive **137** may fill the gap between the first package **140** and the second package **120**. The thermal adhesive **137** may be in direct contact with the solder resist layer **122** of the second package **120**. The opposing surface of the thermal adhesive **137** may be in direct contact with surfaces of the second dies **145** of the first package **140**. Accordingly, the thermal resistance between the first package **140** and the second package **120** is substantially reduced by the presence of the thermal adhesive **137**. In an embodiment, the thermal adhesive **137** conforms to the surface of the barrier **131**. That is,

sidewalls of the thermal adhesive **137** are in direct contact with portions of the barrier **131**. Accordingly, the sidewalls of the thermal adhesive **137** may not be substantially vertical. For example, the sidewalls of the thermal adhesive **137** shown in FIG. **1B** include a curved profile. (28) In an embodiment, the thermal adhesive **137** is entirely surrounded by the barrier **131**. As such, the thermal adhesive **137** is prevented from flowing into the TMI region of the electronic package **100**. This is particularly beneficial, since the presence of the thermal adhesive **137** around the TMIs **143** may result in solder extrusions or other manufacturing defects. Accordingly, containing the flow of the thermal adhesive **137** with the barrier **131** improves the yield.

(29) Referring now to FIG. **1C**, a cross-sectional illustration of an electronic package **100** is shown, in accordance with an additional embodiment. In an embodiment, the electronic package **100** in FIG. **1C** is substantially similar to the electronic package **100** in FIG. **1B**, with the exception that the barrier **131** is oriented in a different direction. Particularly, embodiments may include a barrier **131** with a first surface **135** that is in direct contact with the first electronic package **140**, and a second surface **134** that is in direct contact with the second package **120**. Particularly, the first surface **135** may be substantially flat and be in contact with the first and/or second mold layer **142/148**, and the second surface **134** may be curved or domed and be in contact with the solder resist **122** of the second package **120**. Such a configuration may be indicative of the barrier **131** being printed onto the first package **140**.

(30) Referring now to FIGS. **2A-2C**, plan view illustrations and a cross-sectional illustration of a portion of a top package **220** with a barrier **231** are shown in accordance with various embodiments in order to more clearly illustrate the barrier **231**. Particularly, the illustrations are simplified in order to show only the solder resist layer **222** and the barrier **231**. However, it is to be appreciated that solder balls for the TMI interconnects may surround the barrier **231**. Additionally, while a top package **220** of a PoP architecture is shown, it is to be appreciated that embodiments may also include substantially similar barrier configurations on the bottom package of a PoP architecture.

(31) Referring now to FIG. **2A**, a plan view illustration of the bottom surface of a second package **220** is shown, in accordance with an embodiment. In an embodiment, the bottom surface may include a solder resist layer **222**. A barrier **231** may be disposed over the solder resist layer **222**. As shown, the barrier **231** may form a continuous ring. In the illustrated embodiment, the barrier **231** is shown as having a substantially rectangular shape. However, it is to be appreciated that embodiments may include a barrier **231** with any shape. For example, the barrier **231** may include chamfers or the like to account for any substrate features.

(32) Referring now to FIG. **2B**, a plan view illustration of the bottom surface of a second package **220** is shown, in accordance with an additional embodiment. As shown, a first barrier **231** and a second barrier **232** are provided over the solder resist **222**. The inclusion of a first barrier **231** and a second barrier **232** improves the ability to prevent a thermal adhesive from spreading out towards the TMI regions. For example, if the first barrier **231** is breached by the thermal adhesive, then the second barrier **232** is still present to restrict the flow of the thermal adhesive. In the illustrated embodiment, the first barrier **231** and the second barrier **232** are shown as being substantially concentric with each other. However, it is to be appreciated that the shapes of the first barrier **231** and the second barrier **232** need not be substantially similar, nor does the first barrier **231** need to be substantially centered within the second barrier **232**.

(33) Referring now to FIG. **2C**, a cross-sectional illustration of a portion of a second package **220** is shown, in accordance with an embodiment. FIG. **2C** more clearly illustrates the profile of the barrier **231**. As shown, the barrier **231** may have a first surface **235** that is supported by the solder resist **222** that is over the package substrate **221**. That is, the first surface **235** may be substantially flat. In an embodiment, the second surface **234** of the barrier **231** may be curved or dome shaped. The extent of the curvature of the second surface **234** may be controlled by changing the printing conditions and/or the composition of the material used for the barrier **231**.

(34) In an embodiment, the barrier **231** may have a height **H** and a width **W**. In an embodiment, the

width W may be greater than the height H. However, in other embodiments, the width W may be substantially equal to or less than the height H. In a particular embodiment, a ratio of the width W to the height H (W:H) may be approximately 2:1 or greater, 5:1 or greater, or 10:1 or greater. For example, the width W may be approximately 200 μm and the height H may be approximately 20 μm .

(35) Referring now to FIGS. 3A-3C, a series of cross-sectional illustrations depicting barriers **331** over first packages **340** and second packages **320** is shown, in accordance with an embodiment. FIGS. 3A and 3B illustrate barriers over a second package **320**, and FIG. 3C illustrates a barrier **331** over a first package **340**.

(36) Referring now to FIG. 3A, a cross-sectional illustration of a second (top) package **320** is shown, in accordance with an embodiment. In an embodiment, the second package **320** may be a memory package that comprises one or more memory dies **323**. For example, a pair of memory dies **323** attached together by a DAF **324** is shown in FIG. 3A. The memory dies **323** may be electrically coupled to a package substrate **321** by wire bonds **326**. A mold layer **325** may embed the memory dies **323** and the wire bonds **326**.

(37) In an embodiment, a solder resist **322** is disposed over a backside surface of the package substrate **321** opposite from the mold layer **325**. A plurality of solder balls **327** may pass through the solder resist **322** to contact the package substrate **321**. The solder balls **327** may be positioned along the edges of the package substrate **321**. While a single row of solder balls **327** are shown along each edge, it is to be appreciated that any number of rows of solder balls **327** may be used, depending on the needs of the PoP system. In an embodiment, a flux (not shown) may be applied to the solder balls **327** to aid in the formation of the TMI.

(38) In an embodiment, a barrier **331** may be disposed over the solder resist **322**. The barrier **331** separates an interior surface area of the solder resist **322** from the region with the solder balls **327**. Accordingly, a thermal adhesive (not shown) or the like may be disposed within the barrier **331** without risking the spread of the thermal adhesive around the solder balls **327**. In an embodiment, the barrier **331** also prevents the flux from flowing in towards a center of the second package **320**.

(39) Referring now to FIG. 3B, a cross-sectional illustration of a second (top) package **320** is shown, in accordance with an additional embodiment. In an embodiment, the second package **320** in FIG. 3B is substantially similar to the second package **320** in FIG. 3A, with the exception of there being a first barrier **331** and a second barrier **332**. In an embodiment, the first barrier **331** may be surrounded by the second barrier **332**. Accordingly, there is improved protection against thermal adhesive spreading out towards the solder balls **327**, because there are now two barriers **331** and **332** that confine the thermal adhesive. For example, the second barrier **332** would prevent unwanted flow of a thermal adhesive if the first barrier **331** ever fails.

(40) Referring now to FIG. 3C, a cross-sectional illustration of a first (bottom) package **340** is shown, in accordance with an embodiment. The first package **340** may comprise a package substrate **341** and a plurality of dies **344** and **345** over the package substrate **341**. A mold layer **342** may be disposed over the package substrate **341**, and the mold layer **342** may embed the dies **344** and **345**. In an embodiment, a TMI solder **349** may be disposed in openings through the mold layer **342**.

(41) In an embodiment, the first package **340** may comprise a barrier **331** over the mold layer **342**. In an embodiment, the barrier **331** surrounds the perimeter of the dies **345**. Accordingly, there is a die region inside the perimeter of the barrier **331** and a TMI region with TMI solder **349** outside the perimeter of the barrier **331**. Accordingly, a thermal adhesive (not shown) may be confined to the die region. Furthermore, while a single barrier **331** is shown, it is to be appreciated that a second barrier (similar to the second barrier **332** in FIG. 3B) may also be printed around the perimeter of the barrier **331** to provide additional protection.

(42) In FIGS. 3A-3C, the barriers **331/332** are shown as being printed on either the top package **320** or the bottom package **340**. However, it is to be appreciated that in some embodiments, a first

barrier **331** may be printed on the top package, and a second barrier **332** may be printed on the bottom package. That is, the one or more barriers **331/332** may be printed on multiple surfaces of a PoP architecture.

(43) Referring now to FIGS. **4A-4C**, a series of plan view illustrations of an electronic package **420** is shown, in accordance with various embodiments. While examples of a top package are shown, it is to be appreciated that similar barrier and thermal adhesive configurations may also be implemented on the bottom package.

(44) Referring now to FIG. **4A**, a plan view illustration of the bottom of an electronic package **420** is shown, in accordance with an embodiment. As shown, a plurality of barriers (e.g., first barrier **431**, second barrier **432**, and third barrier **433**) may be disposed over a solder resist layer **422**. In an embodiment, the second barrier **432** and the third barrier **433** may be within a perimeter of the first barrier **431**. In an embodiment, the second barrier **432** and the third barrier **433** may be filled with thermal adhesives **437.sub.A** and **437.sub.B**. Isolating the thermal adhesive **437** to various locations within the first barrier **431** may be used to improve thermal performance. For example, the thermal adhesives **437** may be isolated to hot spot locations of the electronic system. Hot spot locations may include portions of the one or more dies in the bottom package (not shown) that dissipate more heat, (e.g., due to the presence of an increased density of transistors, or the like).

(45) In the illustrated embodiment, a pair of interior barriers (i.e., barrier **432** and barrier **433**) are shown. However, it is to be appreciated that any number of interior barriers may be provided, depending on the needs of the electronic package **420**. Furthermore, while shown as being substantially rectangular, it is to be appreciated that the interior barriers **432/433** may include any shape. In an embodiment, the thermal adhesive **437.sub.A** may have a material composition that is substantially similar to the material composition of the thermal adhesive **437.sub.B**. However, in other embodiments, the thermal adhesives **437.sub.A** and **437.sub.B** may have different material compositions.

(46) Referring now to FIG. **4B**, a plan view illustration of the bottom of an electronic package **420** is shown, in accordance with an additional embodiment. The electronic package **420** in FIG. **4B** may be substantially similar to the electronic package **420** in FIG. **4A**, with the exception that a third thermal adhesive **437.sub.C** fills the remaining portion of the first barrier **431**. In an embodiment, the third thermal adhesive **437.sub.C** may have a different material composition than the first thermal adhesive **437.sub.A** and the second thermal adhesive **437.sub.B**. In a particular embodiment, the first thermal adhesive **437.sub.A** and the second thermal adhesive **437.sub.B** may have a lower thermal resistance than the third thermal adhesive **437.sub.C**.

(47) Referring now to FIG. **4C**, a plan view illustration of the bottom of an electronic package **420** is shown, in accordance with an additional embodiment. The electronic package **420** in FIG. **4C** may be substantially similar to the electronic package **420** in FIG. **4B**, with the exception that a fourth barrier **434** is around a perimeter of the first barrier **431**. The inclusion of a fourth barrier **434** provides additional protection from the thermal adhesives **437** bleeding out into the TMI regions of the electronic package **420**.

(48) Referring now to FIG. **5**, a cross-sectional illustration of an electronic system **570** is shown, in accordance with an embodiment. In an embodiment, the electronic system **570** comprises a board **571**. The board **571** may be a printed circuit board (PCB) or the like. In an embodiment, a PoP package **580** may be electrically coupled to the board **571** with interconnects **575**. For example, the interconnects **575** may be solder balls, or any other suitable interconnect architecture.

(49) In an embodiment, the PoP package **580** comprises a bottom package **540** and a top package **520**. The bottom package comprises a package substrate **541** and a plurality of dies **544/545** attached to the package substrate. A mold layer **542** may embed the plurality of dies **544/545**. In an embodiment, a TMI **543** may pass through the mold layer **542**. The TMI **543** may be electrically coupled to one or more of the dies **544/545** by conductive features (e.g., traces, vias, pads, etc.) in the package substrate (not shown).

(50) In an embodiment, the top package **520** may comprise a package substrate **521** with a solder resist layer **522** over the package substrate **521**. The solder resist layer **522** may have an opening through which the TMI **543** contacts the package substrate **521**. In an embodiment, the top package **520** may comprise one or more dies **523** that are connected to the package substrate **521** by wire bonds **526**. The top package **520** may also comprise a mold layer **525** that embeds the dies **523**.

(51) In an embodiment, a barrier **531** may be disposed between the top package **520** and the bottom package **540**. The barrier **531** may be in direct contact with the solder resist **522** of the top package **520** and with the mold layer **542** of the bottom package **540**. The barrier **531** may be a ring that confines a thermal adhesive **537**. The thermal adhesive **537** may be over the plurality of dies **544/545** of the bottom package **540** in order to improve thermal performance. The barrier **531** prevents the thermal adhesive **537** from flowing into the TMI regions where the TMIs **543** are located.

(52) FIG. **6** illustrates a computing device **600** in accordance with one implementation of the invention. The computing device **600** houses a board **602**. The board **602** may include a number of components, including but not limited to a processor **604** and at least one communication chip **606**. The processor **604** is physically and electrically coupled to the board **602**. In some implementations the at least one communication chip **606** is also physically and electrically coupled to the board **602**. In further implementations, the communication chip **606** is part of the processor **604**.

(53) These other components include, but are not limited to, volatile memory (e.g., DRAM), non-volatile memory (e.g., ROM), flash memory, a graphics processor, a digital signal processor, a crypto processor, a chipset, an antenna, a display, a touchscreen display, a touchscreen controller, a battery, an audio codec, a video codec, a power amplifier, a global positioning system (GPS) device, a compass, an accelerometer, a gyroscope, a speaker, a camera, and a mass storage device (such as hard disk drive, compact disk (CD), digital versatile disk (DVD), and so forth).

(54) The communication chip **606** enables wireless communications for the transfer of data to and from the computing device **600**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a non-solid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not. The communication chip **606** may implement any of a number of wireless standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long term evolution (LTE), Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, Bluetooth, derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The computing device **600** may include a plurality of communication chips **606**. For instance, a first communication chip **606** may be dedicated to shorter range wireless communications such as Wi-Fi and Bluetooth and a second communication chip **606** may be dedicated to longer range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

(55) The processor **604** of the computing device **600** includes an integrated circuit die packaged within the processor **604**. In some implementations of the invention, the integrated circuit die of the processor **604** may be part of an electronic package that comprises a thermal adhesive that is bordered by a barrier, in accordance with embodiments described herein. The term “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory.

(56) The communication chip **606** also includes an integrated circuit die packaged within the communication chip **606**. In accordance with another implementation of the invention, the integrated circuit die of the communication chip **606** may be part of an electronic package that comprises a thermal adhesive that is bordered by a barrier, in accordance with embodiments described herein.

(57) The above description of illustrated implementations of the invention, including what is described in the Abstract, is not intended to be exhaustive or to limit the invention to the precise forms disclosed. While specific implementations of, and examples for, the invention are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize.

(58) These modifications may be made to the invention in light of the above detailed description. The terms used in the following claims should not be construed to limit the invention to the specific implementations disclosed in the specification and the claims. Rather, the scope of the invention is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation. Example 1: an electronic package, comprising: a first package; a second package over the first package and electrically coupled to the first package; and a barrier between the first package and the second package. Example 2: the electronic package of Example 1, wherein the barrier has a flat surface and a domed surface. Example 3: the electronic package of Example 2, wherein the first package is in contact with the domed surface of the barrier, and wherein the second package is in contact with the flat surface of the barrier. Example 4: the electronic package of Example 2, wherein the first package is in contact with the flat surface of the barrier, and wherein the second package is in contact with the domed surface of the barrier. Example 5: the electronic package of Examples 1-4, wherein the barrier is a ring. Example 6: the electronic package of Example 5, wherein an interior of the ring is filled with a thermal adhesive. Example 7: the electronic package of Examples 1-6, further comprising: a second barrier between the first package and the second package. Example 8: the electronic package of Example 7, wherein the second barrier surrounds the barrier. Example 9: the electronic package of Examples 1-8, further comprising: a plurality of barriers between the first package and the second package. Example 10: the electronic package of Example 9, wherein each barrier of the plurality of barriers is a ring, and wherein each ring is filled with a thermal adhesive. Example 11: the electronic package of Example 10, wherein a first thermal adhesive in a first barrier ring is different than a second thermal adhesive in a second barrier ring. Example 12: the electronic package of Examples 1-11, wherein the barrier comprises acrylic polymers and TiO₂. Example 13: the electronic package of Examples 1-12, wherein the first package is a system-on-chip (SoC) package and wherein the second package is a memory package. Example 14: an electronic package, comprising: a first package, wherein the first package comprises: a first package substrate; a first die over the first package substrate; a first mold layer over the first package substrate and around the first die; and a plurality of through mold interconnects (TMIs) through the first mold layer; a second package electrically coupled the first package by the TMIs, wherein the second package comprises: a second package substrate; a second die over the second package substrate; and a solder resist over a surface of the second package substrate opposite from the second die; and a barrier between the first package and the second package. Example 15: the electronic package of Example 14, wherein the barrier is a ring, and wherein the ring is within a boundary defined by the TMIs. Example 16: the electronic package of Example 15, wherein the ring is filled with a thermal adhesive, wherein the thermal adhesive is in contact with the first package and the second package. Example 17: the electronic package of Example 15 or Example 16, further comprising: a second barrier around the barrier. Example 18: the electronic package of Examples 14-17, wherein the barrier has a flat surface and a domed surface. Example 19: the electronic package of Example 18, wherein the first package is in contact with the domed surface of the barrier, and wherein the second package is in contact with the flat surface of the barrier. Example 20: the electronic package of Example 18, wherein the first package is in contact with the flat surface of the barrier, and wherein the second package is in contact with the domed surface of the barrier. Example 21: the electronic package of Examples 14-20, further comprising: a board, wherein the board is electrically coupled to the first package. Example 22: the electronic package, comprising: a package substrate with a first surface and a second surface opposite from the first surface; a plurality of dies attached to the first surface

of the package substrate, wherein the plurality of dies are electrically coupled to the package substrate by wire bonds; a solder resist layer over the second surface of the package substrate; solder balls attached to the package substrate through the solder resist layer; and a barrier ring over the solder resist layer and within a perimeter of the solder balls, wherein the barrier ring comprises a flat surface that is in direct contact with the solder resist layer and a domed surface. Example 23: the electronic package of Example 22, further comprising: a plurality of barrier rings. Example 24: an electronic package, comprising: a package substrate; a first die on the package substrate; a plurality of second dies over the first die; a mold layer over the package substrate, the first die, and the plurality of second dies, wherein backside surfaces of the plurality of second dies are exposed; a plurality of through mold interconnects (TMIs) around the first die; and a barrier ring surrounding the exposed backside surfaces of the plurality of second dies. Example 25: the electronic package of Example 24, further comprising: a plurality of barrier rings.

Claims

1. An electronic package, comprising: a package substrate having a top side and an opposite bottom side; a first die coupled to the top side of the package substrate by first interconnects; a first underfill material around the first interconnects and along a first sidewall and a second sidewall of the first die; a second die over and coupled to the first die by second interconnects, the second die within a footprint of the first die; a third die over and coupled to the first die by third interconnects, the third die laterally spaced apart from the second die, and the third die within the footprint of the first die; a second underfill material around the second interconnects and the third interconnects, the second underfill material laterally between and in contact with the second die and the third die; and a mold layer on the second underfill material, the mold layer having a first sidewall in vertical alignment with the first sidewall of the first die, and the mold layer having a second sidewall in vertical alignment with the second sidewall of the first die.
2. The electronic package of claim 1, further comprising: a structure over the second die and the third die, the structure comprising a metal.
3. The electronic package of claim 2, further comprising: a thermal adhesive between the second die and the structure and between the third die and the structure.
4. The electronic package of claim 1, further comprising: a conductive structure laterally spaced apart from the first die and the second die.
5. The electronic package of claim 1, wherein the first underfill material is only partially along each of the first sidewall and the second sidewall of the first die.
6. The electronic package of claim 1, wherein the second underfill material is only partially along a sidewall of the second die and only partially along a sidewall of the third die.
7. The electronic package of claim 1, wherein the mold layer is laterally between the second die and the third die.
8. A method of fabricating an electronic package, the method comprising: providing a first die; coupling a second die to the first die by first upper interconnects, the second die within a footprint of the first die; coupling a third die to the first die by second upper interconnects, the third die within the footprint of the first die, and the third die laterally spaced apart from the second die; forming an upper underfill material around the first interconnects and the second upper interconnects, the upper underfill material laterally between and in contact with the second die and the third die; forming a mold layer on the upper underfill material, the mold layer having a first sidewall in vertical alignment with a first sidewall of the first die, and the mold layer having a second sidewall in vertical alignment with a second sidewall of the first die coupling the first die to a package substrate with lower interconnects; and forming a lower underfill material around the lower interconnects and along the first sidewall and the second sidewall of the first die.
9. The method of claim 8, further comprising: providing a structure over the second die and the

third die, the structure comprising a metal.

10. The method of claim 9, further comprising: forming a thermal adhesive between the second die and the structure and between the third die and the structure.

11. The method of claim 8, further comprising: forming a conductive structure laterally spaced apart from the first die and the second die.

12. The method of claim 8, wherein the lower underfill material is only partially along each of the first sidewall and the second sidewall of the first die.

13. The method of claim 8, wherein the upper underfill material is only partially along a sidewall of the second die and only partially along a sidewall of the third die.

14. The method of claim 8, wherein the mold layer is laterally between the second die and the third die.
